

Linear Models Lecture 15: GMM Estimation and Efficiency

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Where We Left Off

In Lecture 14, we introduced IV estimation as a **method of moments**:

$$\frac{1}{n} \sum_{i=1}^n Z_i(Y_i - X_i'\beta) = 0$$

This works when the number of moment conditions ℓ equals the number of parameters k .

But what if we have more instruments than parameters?

- We cannot simultaneously set all moment conditions to zero
- We need a principled way to *get as close as possible*
- This is the **overidentification problem**

The Problem

Suppose you have $\ell > k$ moment conditions:

$$\mathbb{E}[g_i(\beta)] = 0, \quad g_i \in \mathbb{R}^\ell, \quad \beta \in \mathbb{R}^k$$

In a sample, you cannot set all ℓ conditions to zero simultaneously—there are more equations than unknowns.

Example: You have three valid instruments for one endogenous variable. Each implies $\mathbb{E}[Z_{ji}e_i] = 0$. No single $\hat{\beta}$ satisfies all three exactly.

We need a principled way to get *as close as possible* to satisfying all of them. The choice of “close” will determine both the estimator and its efficiency.

Moment Equation Models

Let $g_i(\beta)$ be a known $\ell \times 1$ function of the i^{th} observation and a $k \times 1$ parameter β .

Definition

A **moment equation model** is defined by

$$\mathbb{E}[g_i(\beta)] = 0$$

and a parameter space $\beta \in B$.

Identification requires $\ell \geq k$:

- $\ell = k$: **Just identified** — exactly enough information
- $\ell > k$: **Overidentified** — excess information (testable restrictions)
- $\ell < k$: **Underidentified** — insufficient information

Example: IV model: $g_i(\beta) = Z_i(Y_i - X_i'\beta)$

Everything Is a Moment Condition

Estimator	Moment condition $g_i(\beta)$	MME
Sample mean	$Y_i - \mu$	$\hat{\mu} = \bar{Y}$
OLS	$X_i(Y_i - X_i'\beta)$	$\hat{\beta} = (X'X)^{-1}X'Y$
IV	$Z_i(Y_i - X_i'\beta)$	$\hat{\beta} = (Z'X)^{-1}Z'Y$
2SLS	$Z_i(Y_i - X_i'\beta)$ with $W = (Z'Z)^{-1}$	$\hat{\beta}_{2SLS}$
GLS	$\bar{X}_i\Sigma^{-1}(Y_i - \bar{X}_i'\beta)$	$\hat{\beta}_{GLS}$

Key insight: All the estimators we have studied are *method of moments estimators*: they solve $\bar{g}_n(\hat{\beta}) = 0$. GMM generalizes to the overidentified case.

Overidentified IV in the Wild

Each row below has more instruments than endogenous regressors, so no $\hat{\beta}$ can satisfy every sample moment exactly.

Study	Endog. X	Instruments Z (ℓ)	Outcome Y
Angrist (1990)	veteran status	cohort-specific draft eligibility (3–4)	log wages
Miguel et al. (2004)	GDP growth	rainfall growth + 1 lag (2)	civil conflict (Africa)
Dube & Vargas (2013)	commodity income	oil + coffee price shocks $\times$ producer (2)	Colombian conflict
Nunn (2008)	slave exports	distances to 4 trade destinations (4)	log GDP per capita

GMM combines all of Z at once, and the overidentifying restrictions become testable (Lecture 16).

Getting as Close as Possible

Define the sample moment:

$$\bar{g}_n(\beta) = \frac{1}{n} \sum_{i=1}^n g_i(\beta)$$

When $\ell = k$: We can solve $\bar{g}_n(\hat{\beta}) = 0$ exactly.

When $\ell > k$: There are **more equations than unknowns**. In general, no β sets $\bar{g}_n(\beta) = 0$.

The estimator is whatever makes $\bar{g}_n(\beta)$ *small*. “Small” requires a metric on $\mathbb{R}^\ell$, and the metric is what we have to choose.

The GMM Criterion Function

For a positive definite $\ell \times \ell$ weight matrix W , the **GMM criterion function** is:

$$J(\beta) = n \bar{g}_n(\beta)' W \bar{g}_n(\beta)$$

- $J(\beta) \geq 0$ for all β (since $W > 0$)
- When $W = I_\ell$: $J(\beta) = n \|\bar{g}_n(\beta)\|^2$ (Euclidean distance)
- The factor n is for distributional convenience
- Different choices of W yield different estimators with different efficiency

For the linear IV model:

$$J(\beta) = n(Z'Y - Z'X\beta)'W(Z'Y - Z'X\beta)$$

The GMM Estimator

Definition (GMM Estimator)

$$\hat{\beta}_{gmm} = \arg \min_{\beta} J(\beta) = \arg \min_{\beta} n \bar{g}_n(\beta)' W \bar{g}_n(\beta)$$

Theorem (13.1)

For the overidentified linear IV model, the GMM estimator is:

$$\hat{\beta}_{gmm} = (X'ZWZ'X)^{-1}(X'ZWZ'Y)$$

- When $\ell = k$ (just identified): $\hat{\beta}_{gmm} = (Z'X)^{-1}(Z'Y) = \hat{\beta}_{iv}$, regardless of W
- When $\ell > k$: the estimator **depends on W**

2SLS as GMM

Theorem (13.2)

If $W = (Z'Z)^{-1}$, then $\hat{\beta}_{gmm} = \hat{\beta}_{2sls}$.
Furthermore, if $k = \ell$ then $\hat{\beta}_{gmm} = \hat{\beta}_{iv}$.

Proof sketch: Substitute $W = (Z'Z)^{-1}$ into Theorem 13.1:

$$\hat{\beta} = (X'Z(Z'Z)^{-1}Z'X)^{-1}(X'Z(Z'Z)^{-1}Z'Y) = (X'P_ZX)^{-1}(X'P_ZY)$$

where $P_Z = Z(Z'Z)^{-1}Z'$ is the projection matrix — exactly the 2SLS formula.

Takeaway: 2SLS is a *one-step* GMM estimator. But $(Z'Z)^{-1}$ is not always the best weight matrix.

Just-Identified GMM: Mean and Variance

Before tackling sandwich variance, here is GMM solved analytically in a *just-identified* system.

Political-economy setup. For each state i , Y_i is a survey-based measure of state opinion liberalism and X_i contains demographic and partisan covariates. Smaller-sample states yield noisier measurements, so the error variance depends on observables Z_i (intercept and $1/n_i$, where n_i is the respondent count):

$$\begin{aligned} Y_i &= X_i' \beta + e_i, & \mathbb{E}[X_i e_i] &= 0 \\ e_i^2 &= Z_i' \gamma + \eta_i, & \mathbb{E}[Z_i \eta_i] &= 0 \end{aligned}$$

Stack the moment conditions. With parameters $\theta = (\beta', \gamma')'$,

$$g_i(\theta) = \begin{pmatrix} X_i(Y_i - X_i' \beta) \\ Z_i((Y_i - X_i' \beta)^2 - Z_i' \gamma) \end{pmatrix}, \quad \mathbb{E}[g_i(\theta)] = 0.$$

The system is just-identified: $k + \ell$ moments for the $k + \ell$ parameters in θ .

Worked Example: Solving the Sample Analog

Set the sample analog to zero, $\bar{g}_n(\hat{\theta}) = 0$. The two blocks decouple sequentially.

Mean block (top k moments). Setting $\sum_i X_i(Y_i - X_i'\hat{\beta}) = 0$ gives ordinary least squares:

$$\hat{\beta} = (X'X)^{-1}X'Y.$$

Variance block (bottom ℓ moments). Form residuals $\hat{e}_i = Y_i - X_i'\hat{\beta}$. Setting $\sum_i Z_i(\hat{e}_i^2 - Z_i'\hat{\gamma}) = 0$ gives a regression of squared residuals on Z :

$$\hat{\gamma} = (Z'Z)^{-1}Z'\hat{e}^2.$$

$\hat{\gamma}$ is a parametric estimate of the variance function $\text{Var}(e_i | Z_i)$, built on top of $\hat{\beta}$. **How does $\hat{\beta}$'s sampling noise propagate into $\hat{\gamma}$?** The sandwich formula on the next slide answers this for the full vector $\hat{\theta}$.

Asymptotic Distribution of GMM

Let $Q = \mathbb{E}[ZX']$ and $\Omega = \mathbb{E}[ZZ'e^2]$.

Theorem (13.3: Asymptotic Distribution of GMM)

As $n \rightarrow \infty$,

$$\sqrt{n}(\hat{\beta}_{gmm} - \beta) \xrightarrow{d} N(0, V_\beta)$$

where

$$V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}$$

- This is a **sandwich form**: “bread” $(Q'WQ)^{-1}$ wraps “meat” $Q'W\Omega WQ$
- Applies for *any* positive definite weight matrix W
- Different $W \Rightarrow$ different $V_\beta \Rightarrow$ different efficiency

Why Inverse-Variance Weighting?

Toy case: two uncorrelated unbiased estimators of β . Let $\hat{\beta}_1, \hat{\beta}_2$ have asymptotic variances V_1, V_2 . The linear combination $\hat{\beta}_w = w\hat{\beta}_1 + (1 - w)\hat{\beta}_2$ has variance $w^2 V_1 + (1 - w)^2 V_2$, minimized at

$$w^* = \frac{V_2}{V_1 + V_2}.$$

The noisier estimator gets less weight: weight by inverse variance. That is what “as close as possible” means once measurements differ in precision.

The matrix analog. With ℓ moment conditions, $\bar{g}_n(\beta)$ has covariance $\Omega = \mathbb{E}[g_i g_i']$. The natural extension is

$$W = \Omega^{-1}$$

which downweights noisy moments and corrects for correlations. Same idea as GLS, applied to moments rather than observations.

Efficient GMM

Theorem (13.4: Efficient GMM)

Setting $W = \Omega^{-1}$, as $n \rightarrow \infty$: $\sqrt{n}(\hat{\beta}_{gmm} - \beta) \xrightarrow{d} N(0, V_\beta)$ where $V_\beta = (Q'\Omega^{-1}Q)^{-1}$.

The sandwich **collapses**: $(Q'\Omega^{-1}Q)^{-1} \underbrace{(Q'\Omega^{-1}\Omega\Omega^{-1}Q)}_{Q'\Omega^{-1}Q} (Q'\Omega^{-1}Q)^{-1}$

Theorem (13.5: Efficiency)

For any $W > 0$: $(Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1} - (Q'\Omega^{-1}Q)^{-1} \geq 0$.

No GMM estimator with these moment conditions can have a smaller asymptotic variance. Chamberlain (1987): this is the **semiparametric efficiency bound**.

When Is 2SLS Efficient?

Recall 2SLS uses $W = (Z'Z)^{-1}$, which converges to $(\mathbb{E}[ZZ'])^{-1}$.

The efficient weight is $\Omega^{-1} = (\mathbb{E}[ZZ'e^2])^{-1}$.

These are equal when $\mathbb{E}[e^2|Z] = \sigma^2$ (conditional homoskedasticity):

$$\mathbb{E}[ZZ'e^2] = \sigma^2 \mathbb{E}[ZZ']$$

Theorem (13.6)

Under conditional homoskedasticity $\mathbb{E}[e^2|Z] = \sigma^2$, the 2SLS estimator $\hat{\beta}_{2sls}$ is efficient GMM.

Implications:

- Homoskedastic errors $\Rightarrow$ 2SLS is fine, no need for GMM
- Heteroskedastic errors $\Rightarrow$ 2SLS is *inefficient*; use efficient GMM

Two-Step GMM

Problem: $\Omega = \mathbb{E}[ZZ'e^2]$ is unknown. We need to estimate it.

Two-Step GMM:

- Step 1:** Estimate β by 2SLS (using $W = (Z'Z)^{-1}$). Compute residuals $\tilde{e}_i = Y_i - X_i'\hat{\beta}_{2sls}$.
- Step 2:** Estimate $\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n Z_i Z_i' \tilde{e}_i^2$, set $\hat{W} = \hat{\Omega}^{-1}$, and re-estimate β .

Theorem (13.7)

The two-step GMM estimator is asymptotically efficient:

$$\sqrt{n}(\hat{\beta}_{gmm} - \beta) \xrightarrow{d} N(0, V_\beta) \quad \text{where} \quad V_\beta = (Q'\Omega^{-1}Q)^{-1}$$

Key point: The initial estimator does not affect the asymptotic distribution.

Iterated GMM

The two-step estimator's *finite-sample* performance can depend on the initial estimator. To remove this dependence:

Iterated GMM:

- 1 Start with $\hat{\beta}^{(0)}$ (e.g., 2SLS)
- 2 Compute $\hat{\Omega}^{(s)} = \frac{1}{n} \sum Z_i Z_i' (\hat{e}_i^{(s)})^2$
- 3 Re-estimate: $\hat{\beta}^{(s+1)} = (X' Z \hat{\Omega}^{(s)-1} Z' X)^{-1} (X' Z \hat{\Omega}^{(s)-1} Z' Y)$
- 4 Repeat until convergence

In R's `gmm` package: `type = "twoStep"` (two-step) or `type = "iterative"` (iterated).

Note: Hansen and Lee (2021) show the iterated GMM estimator is unaffected by whether the weight matrix is computed with or without centering.

Covariance Matrix Estimation

One-step or two-step GMM (general W):

$$\hat{V}_\beta = (\hat{Q}' \hat{W} \hat{Q})^{-1} (\hat{Q}' \hat{W} \hat{\Omega} \hat{W} \hat{Q}) (\hat{Q}' \hat{W} \hat{Q})^{-1}$$

where $\hat{Q} = \frac{1}{n} \sum Z_i X_i'$ and $\hat{\Omega}$ uses residuals from the final step.

Efficient iterated GMM ($\hat{W} = \hat{\Omega}^{-1}$):

$$\hat{V}_\beta = \left(\hat{Q}' \hat{\Omega}^{-1} \hat{Q} \right)^{-1}$$

The sandwich simplifies; same simplification as GLS vs. OLS.

Practical advice: Always report robust (sandwich) standard errors for one-step GMM. For efficient GMM, the simplified formula is already robust.

Application: Missing Data

Missing data on regressors is pervasive. Abrevaya & Donald (2017) surveyed top journals and found $\sim 50\%$ of empirical papers have missing data; $\sim 70\%$ simply drop incomplete observations.

Standard approaches:

- 1 **Complete case:** Drop missing obs.—consistent, but loses efficiency
- 2 **Dummy variable:** Fill in 0, add indicator—**can be inconsistent**
- 3 **Linear imputation:** Predict missing values—requires extra assumptions

The GMM approach: write down moment conditions for *both* complete and incomplete observations. The complete cases contribute structural moments; the incomplete cases contribute reduced-form moments. GMM combines them optimally.

The Missing Data Model (Abrevaya & Donald, 2017)

Structural regression:

$$y_i = \alpha_0 x_i + z_i' \beta_0 + \varepsilon_i \quad \text{where } E(x_i \varepsilon_i) = 0, E(z_i \varepsilon_i) = 0$$

x_i is a (possibly missing) scalar regressor; z_i is always observed.

Linear projection of x_i on z_i : $x_i = z_i' \gamma_0 + \xi_i$ where $E(z_i \xi_i) = 0$. **Missingness indicator:** $m_i = 1$ if x_i missing, $m_i = 0$ if observed.

Key substitution: For observations with $m_i = 1$:

$$y_i = z_i'(\gamma_0 \alpha_0 + \beta_0) + (\varepsilon_i + \xi_i \alpha_0) \equiv z_i' \delta_0 + \eta_i$$

Assumption 1: (a) $E(m_i z_i \varepsilon_i) = 0$; (b) $E(m_i z_i \xi_i) = 0$; (c) $E(m_i x_i \varepsilon_i) = 0$.
 $\Rightarrow$ Missingness may depend on z_i but not on ε_i or ξ_i .

Three Blocks of Moment Conditions

The parameters are $\theta = (\alpha_0, \beta_0, \gamma_0)'$ with $k = 2K + 1$ unknowns.

Block 1 (observed cases, structural equation): $(1 - m_i) \cdot w_i(y_i - \alpha_0 x_i - z_i' \beta_0) = 0$ $[K + 1$ conditions]

Block 2 (observed cases, projection equation): $(1 - m_i) \cdot z_i(x_i - z_i' \gamma_0) = 0$ $[K$ conditions]

Block 3 (missing cases, substituted equation): $m_i \cdot z_i(y_i - z_i'(\gamma_0 \alpha_0 + \beta_0)) = 0$ $[K$ conditions]

Total: $\ell = 3K + 1$ moments for $k = 2K + 1$ parameters $\Rightarrow K$ **overidentifying restrictions**
(testable via J-test!)

With $K = 2$ (intercept + IQ): 7 moments, 5 parameters, **2 overidentifying restrictions**

Why GMM Beats the Alternatives

Method	Consistent?	Efficient?	Testable?
Complete case	Yes (drops $m_i = 1$)	No (fewer obs)	No
Dummy variable	Sometimes no	No	No
Linear imputation	Yes (under Assn 1)	Somewhat	No
GMM	Yes (under Assn 1)	Yes (optimal W)	Yes (J-test)

Abrevaya & Donald Monte Carlo ($n = 400, 1000$ replications):

Method	Bias (α_0)	$n \times \text{Var} (\alpha_0)$	MSE (α_0)
Complete case	0.011	13.93	0.035
Dummy variable	-0.194	13.94	0.073
FGLS	0.011	13.93	0.035
GMM	0.006	12.53	0.031

WLS Data: Education and BMI

```
library(gmm); library(haven); library(estimatr)

# Wisconsin Longitudinal Study data
data1 <- read_dta("wls-data.dta")
data_men <- subset(data1, male == 1)

# How much data is missing?
table(data_men$bmimissing)
#    0    1
# 4231 1095    (~21% missing)

# Complete-case regression
data_complete <- subset(data_men, bmimissing == 0)
cc_reg <- lm(educ ~ bmirating + iq, data = data_complete)
```

Education (y) regressed on BMI rating (x , sometimes missing) and IQ (z , always observed).

Defining GMM Moment Functions in R

```
g_men <- function(theta, x) {  
  beta_0 <- theta[1]; alpha <- theta[2]; beta_iq <- theta[3]  
  gamma_0 <- theta[4]; gamma_iq <- theta[5]  
  educ <- x[,1]; bmirating <- x[,2]; iq <- x[,3]; bmimissing <- x[,4]  
  
  # Block 1: structural eq (observed cases)  
  r1 <- (1 - bmimissing) * (educ - beta_0 - alpha*bmirating - beta_iq*iq)  
  # Block 2: projection eq (observed cases)  
  r2 <- (1 - bmimissing) * (bmirating - gamma_0 - gamma_iq*iq)  
  # Block 3: substituted eq (missing cases)  
  r3 <- bmimissing * (educ - (gamma_0*alpha+beta_0) - (gamma_iq*alpha+beta_iq)*iq)  
  
  cbind(r1, r1*bmirating, r1*iq, # 3 moments  
        r2, r2*iq, # 2 moments  
        r3, r3*iq) # 2 moments = 7 total, 5 params  
}
```

Running GMM in R

```
# Starting values from complete-case regressions
start_men <- c(beta_0 = coef(cc_reg)[1],
               alpha  = coef(cc_reg)[2],
               beta_iq = coef(cc_reg)[3],
               gamma_0 = coef(proj_reg)[1],
               gamma_iq = coef(proj_reg)[2])

# Two-step GMM with HAC variance
gmm_men <- gmm(g_men, x = x_men, t0 = start_men,
               type = "twoStep",
               wmatrix = "ident", vcov = "HAC")
summary(gmm_men)
```

type = "twoStep": initial $W = I$, then update with $\hat{\Omega}^{-1}$. vcov = "HAC": HAC variance estimator.

Comparing Methods: Results

	Complete Case	Dummy Variable	GMM
$\hat{\alpha}$ (BMI)	−0.0343	−0.0343	−0.0340
SE	(0.0077)	(0.0073)	(0.0072)
$\hat{\beta}_{IQ}$	0.0563	0.0567	0.0564
SE	(0.0017)	(0.0015)	(0.0015)
n used	4231	5326	5326

J-test for overidentifying restrictions:

```
specTest(gmm_men)  # J-test = ?, df = 2, p-value = ?
```

- Large p-value: fail to reject $\Rightarrow$ Assumption 1 is plausible
- Small p-value: evidence against the linear projection restriction

Nonlinear GMM

So far: $g_i(\beta)$ is linear in β . In general, $g_i(\beta)$ can be **nonlinear**.

The GMM estimator still minimizes $J(\beta) = n \bar{g}_n(\beta)' \hat{W} \bar{g}_n(\beta)$ but now requires numerical optimization (no closed form).

Proposition 13.1: Distribution of Nonlinear GMM

$\sqrt{n}(\hat{\beta}_{\text{gmm}} - \beta) \xrightarrow{d} N(0, V_\beta)$ where $V_\beta = (Q'WQ)^{-1}(Q'W\Omega WQ)(Q'WQ)^{-1}$ with the **Jacobian** $Q = \mathbb{E}\left[\frac{\partial}{\partial \beta'} g_i(\beta)\right]$ replacing the linear $\mathbb{E}[ZX']$.

Example: The Abrevaya & Donald missing data model is nonlinear: Block 3 contains $\gamma_0\alpha_0$, a product of parameters.

The MTE Connection

Recall from Lecture 14: the **Marginal Treatment Effect** is

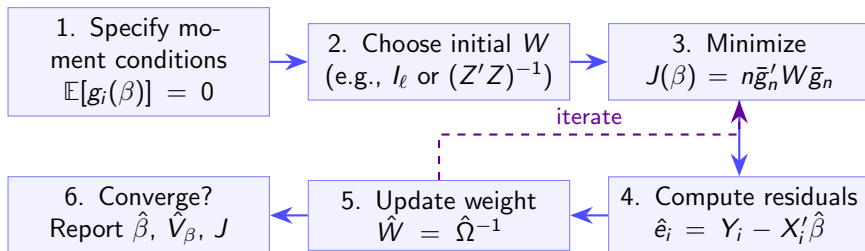
$$\Delta^{MTE}(x, u_D) = \mathbb{E}[Y_1 - Y_0 \mid X = x, U_D = u_D]$$

MTE estimation is nonlinear GMM:

- Moment conditions from the relationship between $E[Y \mid X, Z]$ and propensity score $P(Z)$
- MTE function modeled as a polynomial in u_D ; parameters enter nonlinearly
- Overidentification arises from multiple instruments $\Rightarrow J$ -test applies

The `ivmte` package (Shea & Torgovitsky) implements this: specifying moment conditions, choosing weight matrices, and testing overidentifying restrictions.

Summary of GMM Estimation



- **One-step:** Stop after step 3 (e.g., 2SLS with $W = (Z'Z)^{-1}$)
- **Two-step:** One pass through steps 3–5
- **Iterated:** Repeat steps 3–5 until convergence

The OLS → GLS Parallel

The progression from 2SLS to efficient GMM mirrors Lectures 5–6:

	Linear model	IV model
Baseline	OLS	2SLS
Efficient under homoskedasticity?	Yes	Yes
Efficient under heteroskedasticity?	No $\Rightarrow$ GLS	No $\Rightarrow$ efficient GMM
Variance formula	$(X'\Omega^{-1}X)^{-1}$	$(Q'\Omega^{-1}Q)^{-1}$
Estimated how?	FGLS (two-step)	Two-step GMM

GMM is to IV what GLS is to OLS: use the covariance structure of the moments to improve efficiency.

Where We Stand

What GMM Buys You

- 1 **No distributional assumptions.** You specify moment conditions, not a likelihood—only orthogonality conditions.
- 2 **Optimal weighting.** $W = \Omega^{-1}$ gives the smallest asymptotic variance, paralleling GLS.
- 3 **Testable restrictions.** Extra moment conditions generate overidentifying restrictions you can test with the J -statistic (next lecture).

Every estimator in this course—OLS, GLS, IV, 2SLS—is a special case. GMM makes explicit what each one assumes and what it trades off.

Preview of Next Lecture

Lecture 16: GMM Inference and Going Beyond ATE

- **Testing:** The J-test for overidentification (Thm 13.14)
- **Hypothesis tests:** Wald test and Distance test in the GMM framework
- **Subset tests:** Testing specific instruments, endogeneity tests
- **Application: Propaganda and Foreign Media**
 - Kern & Hainmueller: West German TV in East Germany
 - IV gives LATE ≈ -0.12 (media reduces support for regime)
 - MTE reveals **treatment effect heterogeneity**: always-takers vs. never-takers have *opposite signs*
- **The course in one slide:** OLS $\rightarrow$ GLS $\rightarrow$ IV $\rightarrow$ 2SLS $\rightarrow$ GMM